Software Test Plan (STP) – Online Quiz System

Project: Online Quiz System – console-based C++ MCQ quiz game

Version: 1.1

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Status: Draft (updated to match the implemented repository, test layout and CI)

Changes since v1.0: test levels and tools now match the repository (CTest, GitHub Actions on three operating systems, sanitizers); the "Integration" level is now called "Component"; InputValidation is listed as a module; the schedule and risks are updated for the 31-Oct-2026 deadline; the RTM shows test locations and current status.

1. Introduction

Purpose: This document defines the test plan for the Online Quiz System v1.1. It sets out objectives, scope, strategy, resources, schedule, and responsibilities for testing, derived from the SRS v1.0 (06-09-2026) and the repository's test setup (tests/README.md).

Scope: Testing covers menu navigation and quiz setup, quiz gameplay, scoring and results, leaderboard/history, admin question-bank management, data persistence and error handling, non-functional requirements (performance, reliability, usability, maintainability, portability), and security requirements. GUI, web/mobile, networked/multiplayer play, and LMS integration are out of scope (SRS §1.2).

References: Online Quiz System SRS v1.0; project README and tests/README.md; CI workflow (.github/workflows/ci.yml); Jira project OQSSMP (board 68); ISO/IEC/IEEE 29119 (test documentation).

Definitions: MCQ (Multiple Choice Question), CLI (Command Line Interface), SRS, RTM (Requirements Traceability Matrix), FR/NFR (Functional / Non-Functional Requirement), SR (Security Requirement), AC (Acceptance Criteria), UAT (User Acceptance Testing), CI (Continuous Integration), ASan/UBSan (AddressSanitizer / UndefinedBehaviorSanitizer).

2. Test Items

- MenuModule – main menu, category and difficulty selection
- InputValidation – console input checks (menu choices, answers A–D, skip)
- QuizEngine – question selection, shuffling, presentation, answer validation, skip/timeout
- ScoringModule – score calculation, results summary, grade/remark
- FileIOModule – question bank loading/saving, results persistence, leaderboard and history data
- AdminModule – password setup and authentication, lockout, question add/edit/delete/view and validation
- Build system and CI – CMake build, C++17 on Windows, Linux and macOS

3. Features to be Tested

Features are mapped to SRS requirement IDs. Full test-case mapping is in Section 13.

Area	Requirements	Priority
Quiz setup	QZ-F-001 Main menu; QZ-F-002 Category selection; QZ-F-003 Difficulty selection; QZ-F-004 Random selection, no repeats	High / Medium
Quiz gameplay	QZ-F-010 Four-option MCQ display; QZ-F-011 Option shuffling; QZ-F-012 A–D input with re-prompt; QZ-F-013 Optional time limit; QZ-F-014 Skip; QZ-F-015 Immediate feedback	High / Medium / Low

Area	Requirements	Priority
Scoring & results	QZ-F-020 Score; QZ-F-021 Results summary; QZ-F-022 Grade/remark; QZ-F-023 Persist attempt; QZ-F-024 Leaderboard; QZ-F-025 Player history	High / Medium
Admin	QZ-F-030 Password-protected admin menu; QZ-F-031 Add; QZ-F-032 Edit/delete by ID; QZ-F-033 View/filter; QZ-F-034 Question validation	High / Medium
Data & errors	QZ-F-040 Load bank, clear error on missing/corrupt file; QZ-F-041 Immediate persistence; QZ-F-042 Invalid menu input handling	High
Non-functional	QZ-NF-001 Start-to-first-question ≤ 2 s; QZ-NF-002 No data loss on abrupt termination; QZ-NF-003 Usability; QZ-NF-004 Modularity; QZ-NF-005 Cross-platform build; QZ-NF-006 Data-driven categories	High / Medium / Low
Security	QZ-SR-001 Admin password required; SR-002 Answer hidden pre-submit; SR-003 Password not stored in plaintext; SR-004 File input validation; SR-005 No player-facing write path; SR-006 Lockout after 3 failed attempts	High / Medium

4. Features Not to be Tested

- GUI, web/mobile interfaces, networked or multiplayer play, LMS integration (out of scope per SRS)
- C++ standard library internals and compiler behaviour
- Terminal emulator behaviour beyond basic text input and output
- Concurrent multi-user access (system is single-user per session)
- Cloud/shared-file export of results (future extension, not in v1.0)
- Hiding of typed password characters (not portable across operating systems; documented limitation)

5. Test Approach / Strategy

Levels (matching the repository layout; CMake discovers all tests automatically):

- **Unit tests (tests/unit/test_<module>.cpp):** one module on its own – scoring logic, input validation, question validation, shuffling, file parsing and saving, admin authentication and CRUD.
- **Component tests (tests/component/test_<module>.cpp):** modules working together – e.g. MenuModule with InputValidation, QuizEngine with ScoringModule and FileIOModule, AdminModule with FileIOModule (edits reflected in the next quiz). This level was called "Integration" in v1.0.
- **System tests (tests/system/<name>.in + <name>.expect):** the real quiz program is run with scripted keyboard input and its output is checked line by line – start-up, menu navigation, invalid input, closed input, quiz setup, results menu, full quiz flow and admin flow.
- **Acceptance tests (UAT):** execution of the SRS acceptance suites – Menu Navigation, Quiz Gameplay, Scoring & Results, Admin Question Management, Data Persistence, Security, Usability – by a team member who did not write the code.

Types:

- Functional testing (all QZ-F requirements)
- Regression testing (the full CTest suite runs on every pull request and on every push to main)
- Portability testing (CI matrix: Ubuntu, Windows, macOS – QZ-NF-005)
- Memory and undefined-behaviour checking (separate CI job running every test under ASan and UBSan on Linux)

- Reliability testing (corrupt/missing files, failed file replacement, invalid and closed input)
- Performance testing (start-up to first question – QZ-NF-001)
- Usability walkthrough with a first-time user (QZ-NF-003)
- Boundary and negative testing (invalid menu choices, empty options, impossible dates, out-of-range scores, oversized input)

Entry Criteria: The branch builds cleanly with CMake on all three operating systems; test data files are available; the pull request is open and CI has started.

Exit Criteria: All planned unit, component and system tests pass in CI on all three operating systems and under sanitizers; all high-priority functional requirements implemented and verified; no critical NFR failures; 0 critical/high defects open; the RTM shows all test cases passed (SRS §6); every pull request merged with green CI and at least one teammate's approval.

5.1 Security Validation

- Admin actions blocked without a correct password (TC-Sec-01) – verified in unit tests
- Correct answer never shown before submit/skip (TC-Sec-02) – system test on the quiz flow
- Admin password stored only as a salted, iterated SHA-256 hash; not recoverable from stored files or source (TC-Sec-03)
- Malformed question and results records are skipped with a warning and never crash the program (TC-Sec-04)
- Code review of every pull request confirms no player-facing path edits questions or scores (TC-Sec-05)
- Fourth consecutive admin login attempt is blocked until restart, even with the correct password (TC-Sec-06)
- Negative and malformed-input testing of console input and of the question/results files (empty and oversized fields, delimiter and line-break characters, impossible dates)
- Sanitizer job (ASan/UBSan) and a read-only CI token as additional safeguards

6. Test Environment

Hardware: Developer laptops (keyboard and terminal) and GitHub-hosted CI runners (ubuntu-latest, windows-latest, macos-latest).

Software: CMake and CTest build of the quiz executable (C++17); plain g++ -std=c++17 for single-module checks; Git and GitHub with pull-request review; MSYS2 g++ on Windows developer machines.

Tools: CTest with the repository's test helpers (unit and component tests), CMake script runner for system tests (.in/.expect files), GitHub Actions (build, test, sanitizers, JUnit reports, downloadable builds), AddressSanitizer and UndefinedBehaviorSanitizer, Jira (project OQSSMP) for tasks, and the RTM sheet (OQSSMP-27).

Test Data: data/questions.txt with 90 questions (10 for each of three categories × three difficulty levels); edge-case question and results files created by the tests at run time (empty, missing, corrupt, duplicate IDs, empty option text, no/multiple correct answers, impossible dates); scripted input files for system tests.

7. Test Schedule

Dates are tentative and tied to the 31-Oct-2026 submission deadline.

Milestones:

- Test plan v1.1 review and approval: 27-Oct-2026
- Module pull requests merged (quiz engine, scoring, admin) with unit tests: by 28-Oct-2026
- Integration of all modules in main.cpp and component tests: 28-Oct-2026 to 29-Oct-2026
- System tests and regression run on all three operating systems: 29-Oct-2026
- RTM update and defect fixes: 29-Oct-2026 to 30-Oct-2026
- UAT: 30-Oct-2026

- Test Summary Report and final submission: 30-Oct-2026 to 31-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test source files in tests/unit, tests/component and tests/system
- Test data (data/questions.txt and test-generated edge-case files)
- CI results and JUnit test reports (GitHub Actions)
- Defect reports and task tracking (Jira, GitHub pull-request comments)
- Updated RTM
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA & Documentation Lead	Ashvin Ronald Lobo	Maintain test plan and RTM, validation and scoring tests, execute UAT, test summary report
Team Lead / Coordinator	Chiranth R.	Repo structure, CMake and CI, MenuModule and InputValidation tests, integration, release readiness
Developer – Quiz Engine & Scoring	Abdul Mateen Shaikh	Unit and component tests for engine and scoring; fix defects in those modules
Developer – Question Bank & Admin	Tanvi B. Shetty	Unit tests for FileIOModule and AdminModule; question bank data; fix defects in those modules
Course Coordinator	<Name>	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Short remaining time; integration depends on the quiz engine, scoring and admin pull requests	Merge in dependency order; keep pull requests small; review within hours; integration branch starts as soon as the modules are merged
Cross-platform differences (line endings, file replacement, console input, timers) break portability or the optional time limit (QZ-F-013)	CI matrix on Linux, Windows and macOS runs every test on every pull request; time limit stays behind a feature flag
Tests that rely on assert() are skipped in Release builds	CI builds in Debug; new tests use the repository's check macros as described in tests/README.md
Corrupt or hand-edited data files crashing the program	Dedicated malformed-file unit tests (QZ-SR-004, QZ-F-040); validation before every save
Weak or missing review of a pull request	At least one teammate's approval and green CI required before merge; reviewers build and run the code
SRS ambiguities (e.g. score bands, leaderboard size N)	Decisions recorded in the RTM comments and in the README

11. Assumptions & Dependencies

- Module names and interfaces follow SRS v1.0 and the component architecture diagram (docs/UML Component Diagram.pdf); InputValidation was added as a separate module during implementation
- CI runs in Debug configuration so assert-based tests are active
- Score bands for grade/remark (QZ-F-022) are defined before the scoring tests are finalised

- Jira board OQSSMP and GitHub pull requests are the source of truth for task and review status
- data/admin.cfg is created at first run and is not part of the repository

12. Suspension & Resumption Criteria

Suspend testing if:

- CI on main is red or the build fails to compile
- The build is too unstable (blocks > 30% of test cases)
- GitHub Actions or the shared repository is unavailable for > 4 hours

Resume testing if:

- Blocking defects are fixed and CI is green again
- The CI service or repository is available

13. Test Case Management & Traceability

The RTM maps every SRS requirement to its test case and to where that test lives. Status legend: N = Not started, P = Partial (implemented or merged, verification incomplete), A = Achieved (test passing in CI). Status shown is as of 27-Oct-2026; the QA lead maintains the live version (OQSSMP-27). "planned" means the test file does not exist yet.

Req ID	Requirement	Priority	Test Case	Test location	St.
QZ-F-001	Main menu with 4 options	High	TC-Menu-01	component: test_menu; system: menu_exit	P
QZ-F-002	Category selection	High	TC-Menu-02	component: test_menu; system: quiz_setup	P
QZ-F-003	Difficulty selection	Medium	TC-Menu-03	component: test_menu; system: quiz_setup	P
QZ-F-004	Random selection, no duplicates	High	TC-Menu-04	unit: test_quiz_engine (planned)	N
QZ-F-010	MCQ with options A–D	High	TC-Quiz-01	unit: test_quiz_engine (planned)	N
QZ-F-011	Shuffle answer options	Medium	TC-Quiz-02	unit: test_quiz_engine (planned)	N
QZ-F-012	A–D input, re-prompt on invalid	High	TC-Quiz-03	unit: test_input_validation; system: menu_invalid_input	P
QZ-F-013	Optional per-question time limit	Low	TC-Quiz-04	unit: test_quiz_engine (planned)	N
QZ-F-014	Skip question	Medium	TC-Quiz-05	unit: test_quiz_engine (planned)	N
QZ-F-015	Immediate feedback	Low	TC-Quiz-06	unit: test_quiz_engine (planned)	N
QZ-F-020	Score calculation	High	TC-Score-01	unit: test_scoring (planned)	N
QZ-F-021	Results summary	High	TC-Score-02	unit: test_scoring (planned)	N
QZ-F-022	Grade/remark	Medium	TC-Score-03	unit: test_scoring (planned)	N
QZ-F-023	Persist attempt	High	TC-Score-04	unit: test_fileio	A
QZ-F-024	Leaderboard	Medium	TC-Score-05	component (planned)	N
QZ-F-025	Player history	Medium	TC-Score-06	component: results_menu (planned)	N
QZ-F-030	Admin password protection	High	TC-Admin-01	unit: test_admin	P
QZ-F-031	Add question	High	TC-Admin-02	unit: test_admin	P
QZ-F-032	Edit/delete by ID	High	TC-Admin-03	unit: test_admin	P
QZ-F-033	View/filter questions	Medium	TC-Admin-04	unit: test_admin	P

Req ID	Requirement	Priority	Test Case	Test location	St.
QZ-F-034	Validate question data	High	TC-Admin-05	unit: test_fileio, test_admin	A
QZ-F-040	Load bank; error on missing/corrupt	High	TC-Data-01	unit: test_fileio	A
QZ-F-041	Persist edits immediately	High	TC-Data-02	unit: test_fileio, test_admin	A
QZ-F-042	Handle invalid menu input	High	TC-Data-03	unit: test_input_validation; system: menu_invalid_input, input_closed	P
QZ-NF-001	Startup ≤ 2 s	High	TC-Perf-01	system: timed start-up (planned)	N
QZ-NF-002	No data loss on termination	High	TC-Rel-01	unit: test_fileio (failed replace keeps original); manual kill test (planned)	P
QZ-NF-003	Usability for first-time user	Medium	TC-UX-01	manual walkthrough (UAT)	N
QZ-NF-004	Modular code structure	Medium	TC-Maint-01	code review of every pull request	P
QZ-NF-005	Windows/Linux/macOS build	Medium	TC-Port-01	CI matrix (all tests, three operating systems)	A
QZ-NF-006	Add categories via data only	Low	TC-Maint-02	unit: test_fileio (planned case)	N
QZ-SR-001	Admin password required	High	TC-Sec-01	unit: test_admin	P
QZ-SR-002	Hide answer before submit	High	TC-Sec-02	system: quiz flow (planned)	N
QZ-SR-003	Password stored hashed	Medium	TC-Sec-03	unit: test_admin	P
QZ-SR-004	Validate/sanitize file inputs	High	TC-Sec-04	unit: test_fileio	A
QZ-SR-005	No player-facing write path	High	TC-Sec-05	code review of every pull request	P
QZ-SR-006	Lockout after 3 failed attempts	Medium	TC-Sec-06	unit: test_admin; system: admin flow (planned)	P

Key test scenarios (examples)

Test Case	Scenario / Expected result	Requirement
TC-Menu-04	Start a 10-question quiz repeatedly; no question repeats within one attempt	QZ-F-004
TC-Quiz-03	Enter E, 1, blank or ab at the answer prompt; the system re-prompts and the score is unchanged	QZ-F-012
TC-Score-02	Mix of correct, incorrect and skipped answers; correct + incorrect + skipped = total; percentage matches	QZ-F-021
TC-Admin-05	Add a question with no correct answer, an empty option, or a | or line break in the text; it is rejected with a clear error and the file is unchanged	QZ-F-034
TC-Data-01	Delete or corrupt the questions file and start the program; readable error, no crash	QZ-F-040
TC-Rel-01	Simulate a failed file replacement and kill the process mid-edit; previously saved data intact	QZ-NF-002

Test Case	Scenario / Expected result	Requirement
TC-Sec-04	Results line with score above total, impossible date such as 31 Feb, or malformed fields; skipped with a warning, rest of the file still loads	QZ-SR-004
TC-Sec-06	Enter a wrong admin password three times, then the correct one; the fourth attempt is blocked until restart	QZ-SR-006

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed and % passed / failed (from CTest and JUnit reports)
- CI pass rate per operating system and for the sanitizer job
- Defect count and severity per module, and defect aging (Jira and pull-request comments)
- Requirement coverage (requirements with at least one passing test, from the RTM)
- Pull-request review coverage (every merged pull request has an approval and green CI)

Reports:

- CI status on every pull request and on main
- Updated RTM
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Ashvin Ronald Lobo	
Team Lead	Chiranth R.	
Course Coordinator		